

Hochschild cohomology of graded categories

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1 Preliminaries on Hochschild cohomology

Definition 1.1. A differential graded category (DG category) is a category with each Hom-set being a chain complex such that

- The composition maps $\text{Hom}(B, C) \otimes \text{Hom}(A, B) \rightarrow \text{Hom}(A, C)$ for all $A, B, C \in \mathcal{C}$ is a map of complexes
- $d(id_A) = 0$ for all $A \in \mathcal{C}$.

Similarly, one can define a graded category as a category in which each Hom-set is a graded R -module for some ring R .

Let $\mathcal{C}$ be a graded category. We denote the category of graded modules over the field k as $\mathbf{GMod}_k$.

Definition 1.2. Let $\mathcal{C}$ be a graded category. A left $\mathcal{C}$ -module is a functor $\mathcal{C} \rightarrow \mathbf{GMod}_k$. A right $\mathcal{C}$ -module is a functor $\mathcal{C}^{op} \rightarrow \mathbf{GMod}_k$. A $\mathcal{C}$ -bimodule is a functor $\mathcal{C}^{op} \otimes \mathcal{C} \rightarrow \mathbf{GMod}_k$.

Example 1.3. We view $\mathcal{C}$ as a $\mathcal{C}$ -bimodule, which is defined as $\mathcal{C}^{op} \times \mathcal{C} \rightarrow \mathbf{GMod}_k$, $\mathcal{C}(x, y) = \text{Hom}_{\mathcal{C}}(x, y)$.

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Now we plan to construct the Hochschild cohomology of the graded category $\mathcal{C}$ with coefficients in the $\mathcal{C}$ -bimodule M . The difference between Hochschild cohomology of a category and the ordinary singular cohomology of a topological space is that the former is bigraded. We will see that one grading comes from the n -th cohomology of the cochain complex, called the Hochschild degree or horizontal degree; and the other grading comes from the coefficients $\mathcal{C}$ -bimodule M .

The following constructs a bigraded cochain complex $C^{\star,*}(\mathcal{C}, M)$.

Construction 1.4. We use $\star$ and $*$ to denote the two degrees of the cochain complex, respectively, as $C^{\star,*}(\mathcal{C}, M)$.

We define $C^{n,*}(\mathcal{C}, M) = \coprod_{X_0, X_1, \dots, X_n} \text{Hom}_k^*(\mathcal{C}(X_1, X_0) \otimes \mathcal{C}(X_2, X_1) \otimes \dots \otimes \mathcal{C}(X_n, X_{n-1}), M(X_n, X_0))$. Here $M(X_n, X_0) \in \mathbf{GMod}_k$ is a graded k -module, and the notation Hom_k^d means the maps of degree d which maps the i^{th} component of domain to $i + d^{\text{th}}$ component of codomain.

There is a bidegree $(1, 0)$ differential

$$\begin{aligned} d(\varphi)(f_1, \dots, f_{n+1}) &= (-1)^{|\varphi|_v|f_1|} f_1 \cdot \varphi(f_2, \dots, f_{n+1}) \\ &+ \sum_{i=1}^n (-1)^i \varphi(\dots, f_i f_{i+1}, \dots) + (-1)^{n+1} \varphi(f_1, \dots, f_n) \cdot f_{n+1}. \end{aligned}$$

Remark 1.5. This is quite similar to the definition of degenerate maps of the nerve of a category.

Here We have $\varphi \in C^{n,*}(\mathcal{C}, M)$ is of the form

$$(\varphi_{X_0, \dots, X_n})_{(X_0, \dots, X_n) \in \mathcal{C}^{n+1}},$$

where $\varphi_{X_0, \dots, X_n} \in \text{Hom}(\mathcal{C}(X_1, X_0) \otimes \dots \otimes \mathcal{C}(X_n, X_{n-1}), M(X_n, X_0))$. So the above formula gives a “rule” for computing

$$(d\varphi)_{(X_0, \dots, X_{n+1}) \in \mathcal{C}^{n+2}}$$

by fixing an index $(X_0, \dots, X_{n+1}) \in \mathcal{C}^{n+2}$.

A detailed formula is:

$$\begin{aligned} (d\varphi)_{X_0, \dots, X_{n+1}}(f_1, \dots, f_{n+1}) &= (-1)^{|\varphi|_v|f_1|} f_1 \cdot \varphi_{X_0, \dots, X_{n+1}}(f_2, \dots, f_{n+1}) \\ &+ \sum_{i=1}^n (-1)^i \varphi_{X_0, \dots, X_i, \dots, X_{n+1}}(\dots, f_i f_{i+1}, \dots) \\ &+ (-1)^{n+1} \varphi_{X_0, \dots, X_n}(f_1, \dots, f_n) \cdot f_{n+1}. \end{aligned}$$

Here we have a composable string

$$X_0 \xleftarrow{f_1} X_1 \xleftarrow{f_2} X_2 \leftarrow \dots \xleftarrow{f_n} X_n \xleftarrow{f_{n+1}} X_{n+1},$$

with

$$\varphi(f_2, \dots, f_{n+1}) \in M(X_{n+1}, X_1), \quad f_1 \in \mathcal{C}(X_1, X_0).$$

$f_1 \cdot \phi(f_2, \dots, f_{n+1})$ is obtained from left action:

Let $M: \mathcal{C}^{\text{op}} \otimes \mathcal{C} \rightarrow \mathbf{GMod}_k$ be a bimodule.

Let $f: X' \rightarrow X$ be a morphism in $\mathcal{C}$. There is a right action denoted by $- \circ f$,

$$- \circ f: M(X, Y) \longrightarrow M(X', Y),$$

induced by the functoriality of $M: \mathcal{C}^{\text{op}} \otimes \mathcal{C} \rightarrow \mathbf{GMod}_k$.

$$\begin{array}{ccc} (X, Y) & \longmapsto & M(X, Y) \\ (f, id) \downarrow & & \downarrow - \circ f \\ (X', Y) & \longmapsto & M(X', Y) \end{array}$$

Similarly, for a morphism $g: Y \rightarrow Y'$ in $\mathcal{C}$, we have a left action denoted by

$$g \circ -: M(X, Y) \longrightarrow M(X, Y').$$

Some useful operations on Hochschild cohomology More details can be found in [Section1, [3]]

- Cup product

$$\smile: C^{p,q}(\mathcal{C}; M) \times C^{s,t}(\mathcal{C}; N) \longrightarrow C^{p+s, q+t}(\mathcal{C}; M \otimes_{\mathcal{C}} N)$$

defined by

$$(\varphi \smile \psi)(f_1, \dots, f_{p+s}) = (-1)^{t \sum_{i=1}^p |f_i|} \varphi(f_1, \dots, f_p) \otimes \psi(f_{p+1}, \dots, f_{p+s})$$

- Note that $\phi \smile \psi = (-1)^{|\phi|_h |\psi|_h} (-1)^{|\phi|_v |\psi|_v} \psi \smile \phi$. Let $d' \phi := (-1)^{|\phi|_v} d \phi$ and $\phi \cdot \psi := (-1)^{|\phi|_h |\psi|_v} \psi \cdot \phi$ and then we have $\phi \cdot \psi = (-1)^{|\phi| |\psi|} \psi \cdot \phi$ and then $HH^{*,*}(\mathcal{C}, \mathcal{C})$ is graded commutative. It is always important for the DG-structure to appear on the cochain level, see the discussion in [Section 0, [2]]

- Pre-Lie product $\bullet: C^{p,q}(\mathcal{C}; \mathcal{C}) \otimes C^{s,t}(\mathcal{C}; \mathcal{C}) \longrightarrow C^{p+s-1, q+t}(\mathcal{C}; \mathcal{C})$

$$\begin{aligned} (\varphi \bullet \psi)(f_1, \dots, f_{p+s-1}) &= \sum_{i=1}^p (-1)^{(s-1)(p-i)+t(p-1+\sum_{j=1}^{i-1} |f_j|)} \\ &\quad \varphi(f_1, \dots, f_{i-1}, \psi(f_i, \dots, f_{i+s-1}), f_{i+s}, \dots, f_{p+s-1}) \end{aligned}$$

Omit the sign; it is like a composition in operads.

- Gerstenhaber bracket

$$[-, -]: C^{p,q}(\mathcal{C}; \mathcal{C}) \otimes C^{s,t}(\mathcal{C}; \mathcal{C}) \longrightarrow C^{p+s-1, q+t}(\mathcal{C}; \mathcal{C})$$

defined by

$$[\varphi, \psi] = \varphi \bullet \psi - (-1)^{(p+q-1)(s+t-1)} \psi \bullet \varphi$$

. Hence, $[\varphi, \psi] = \varphi \bullet \psi - \psi \bullet \varphi$ or $\varphi \bullet \psi + \psi \bullet \varphi$

- Gerstenhaber square

$$\text{Sq} : HH^{p,q}(\mathcal{C}; \mathcal{C}) \longrightarrow HH^{2p-1, 2q}(\mathcal{C}; \mathcal{C})$$

is induced by

$$\text{Sq}(\varphi) = \varphi \bullet \varphi$$

There are many relations among the operations known as the Gerstenhaber square, which can be found in [Section 1, [3]].

Example 1.6. (Differentiate is equivalent to apply bracket) **Claim:** $d' = [m_2, -]$, where $m_2 \in C^{2,0}(\mathcal{C}, \mathcal{C})$ is the composition. More explicitly, consider composition

$$\circ : C(X_1, X_0) \otimes C(X_2, X_1) \longrightarrow C(X_2, X_0)$$

defined by $f \otimes g \longmapsto f \circ g$. It lies in $\text{Hom}_k(\mathcal{C}(X_1, X_0) \otimes \mathcal{C}(X_2, X_1), \mathcal{C}(X_2, X_0))$. We denote m_2 be the image of $\circ$ in $C^{2,0}(C, C)$.

Let $s = |\varphi|_h$, $t = |\varphi|_v$, then

$$\begin{aligned} [m_2, \varphi](f_1, \dots, f_{s+1}) &= m_2 \bullet \varphi(f_1, \dots, f_{s+1}) \\ &\quad - (-1)^{s+t-1} \varphi \bullet m_2(f_1, \dots, f_{s+1}) \end{aligned} \quad (*)$$

$$\begin{aligned} m_2 \bullet \varphi(f_1, \dots, f_{s+1}) &= (-1)^{s-1} m_2(\varphi(f_1, \dots, f_s), f_{s+1}) \\ &\quad + (-1)^{t(1+|f_1|)} m_2(f_1, \varphi(f_2, \dots, f_{s+1})). \end{aligned}$$

$$\begin{aligned} \varphi \bullet m_2(f_1, \dots, f_{s+1}) &= \sum_{i=1}^s (-1)^{s-i} \varphi(f_1, \dots, f_{i-1}, \\ &\quad m_2(f_i, f_{i+1}), f_{i+2}, \dots, f_{s+1}). \end{aligned}$$

Hence,

$$\begin{aligned} (*) &= (-1)^{s-1+t} \varphi(f_1, \dots, f_s) f_{s+1} \\ &\quad + (-1)^{t(1+|f_1|)} f_1 \cdot \varphi(f_2, \dots, f_{s+1}) \\ &\quad - \sum_{i=1}^s (-1)^{t-1-i} \varphi(\dots, f_i f_{i+1}, \dots, f_{s+1}). \end{aligned}$$

On the other hand,

$$\begin{aligned} d' \varphi(f_1, \dots, f_{s+1}) &= (-1)^t d \varphi(f_1, \dots, f_{s+1}) \\ &= (-1)^{t(|f_1|+1)} f_1 \cdot \varphi(f_2, \dots, f_{s+1}) \\ &\quad + \sum_{i=1}^s (-1)^{i+t} \varphi(\dots, f_i f_{i+1}, \dots) \\ &\quad + (-1)^{s+t+1} \varphi(f_1, \dots, f_s) f_{s+1} \\ &= (*). \end{aligned}$$

Thus $d' = [m_2, -]$.

Finally, it comes to the functoriality of Hochschild cohomology.

Construction 1.7. Let $F : \mathcal{D} \rightarrow \mathcal{C}$ be a graded functor between graded categories and $\tau : N \rightarrow M$ be the $\mathcal{C}$ -bimodule map. We define

$$F^* : C^{*,*}(\mathcal{C}; M) \longrightarrow C^{*,*}(\mathcal{D}; M(F, F)), \text{ defined by } F^*(\varphi)(g_1, \dots, g_n) \\ = \varphi(F(g_1), \dots, F(g_n));$$

$$\tau^* : C^{*,*}(\mathcal{C}; N) \longrightarrow C^{*,*+|\tau|}(\mathcal{C}; M), \\ \tau_*(\varphi)(f_1, \dots, f_n) = \tau(\varphi(f_1, \dots, f_n)).$$

Both are obvious constructions, and it can be proved that they induce maps on Hochschild cohomology.

2 Example: Construct a graded category from an ungraded category together with an automorphism and the Hochschild cohomology

In this section, we focus on grading. An important observation is that we can always view an ungraded object as a graded object concentrated at the zero degree. For example, an ungraded category is a graded category concentrated at the degree zero; a module is a chain complex concentrated at the zero component, which is helpful in projective/injective resolution, when $\mathcal{C}$ is an ungraded category, $HH^p, q(\mathcal{C}, M) = HH^p(\mathcal{C}, M^q)$ where $\mathcal{C}$ is viewed as a graded and ungraded category on the left-hand side and right-hand side, respectively.

$$C^{p,q}(\mathcal{C}, M) = \coprod_{X_0, \dots, X_p \in \mathcal{C}} \text{Hom}_k^q(\mathcal{C}(X_1, X_0) \otimes \dots \otimes \mathcal{C}(X_p, X_{p-1}), M(X_p, X_0)) \\ \cong \coprod_{X_0, \dots, X_p \in \mathcal{C}} \text{Hom}_k(\mathcal{C}(X_1, X_0) \otimes \dots \otimes \mathcal{C}(X_p, X_{p-1}), M^q(X_p, X_0)) \\ =: C^p(\mathcal{C}, M^q)$$

The last equation is because $\mathcal{C}(X_1, X_0) \otimes \dots \otimes \mathcal{C}(X_p, X_{p-1})$ has degree 0, and thus any map of degree q lands in M^q . Taking cohomology, we obtain $HH^{p,q}(\mathcal{C}, M) = HH^p(\mathcal{C}, M^q)$.

Construction 2.1. Given an ungraded category $\mathcal{J}$ and an automorphism $\Sigma : \mathcal{J} \rightarrow \mathcal{J}$, we can construct a graded category $\mathcal{J}_\Sigma$ as follows.

- $ob(\mathcal{J}_\Sigma) = ob(\mathcal{J})$
- $\mathcal{J}_\Sigma^n(X, Y) := \mathcal{J}(X, \Sigma^n Y)$ for $n \in \mathbb{Z}$ (Note that $n \in \mathbb{Z}$, rather than $n \in \mathbb{N}$. This leads to a difference between Tate Ext and ordinary Ext)

- The composition is defined as $\mathcal{J}_\Sigma^p(Y, Z) \otimes \mathcal{J}_\Sigma^q(X, Y) \rightarrow \mathcal{J}_\Sigma^{p+q}(X, Z)$, $f \otimes g \mapsto (\Sigma^q f)g$. Things get clear when we translate language into $\mathcal{J}$. $f \in \mathcal{J}_\Sigma^p(Y, Z) = \mathcal{J}(Y, \Sigma^p Z)$. So $\Sigma^q f \in \Sigma^q \mathcal{J}(Y, \Sigma^p Z) = \mathcal{J}(\Sigma^q Y, \Sigma^{p+q} Z)$. So it makes sense to composite $\Sigma^q f$ and $g \in \mathcal{J}(X, \Sigma^q Y)$.

Definition 2.2. A graded category equivalent to $\mathcal{J}_\Sigma$ for some $(\mathcal{J}, \Sigma)$ is called a weakly stable category.

Remark 2.3. We can extend $\Sigma : \mathcal{J} \rightarrow \mathcal{J}$ to an automorphism on $\mathcal{J}_\Sigma$ by $(-1)^n \Sigma$.

Remark 2.4. There is a natural isomorphism of degree -1 given by $id_X, \iota : X \simeq \Sigma X$. Note that $id_X \in \mathcal{J}(X, X) = \mathcal{J}_\Sigma^{-1}(X, \Sigma X)$, so ι has degree -1 , not 0 .

Example 2.5. Algebra Λ can be viewed as a category with one object. Let $\sigma \in \Lambda$. We can construct a graded algebra Λ .

There is a long exact sequence about $\mathrm{HH}^{*,*}(\mathcal{J}_\Sigma, \mathcal{J}_\Sigma)$ and $\mathrm{HH}^{*,*}(\mathcal{J}, \mathcal{J}_\Sigma)$, see [Proposition 4.3, [3]].

3 Hochschild-Tate cohomology

Let Λ be a Frobenius algebra. The most important feature we use in these notes is as follows.

Proposition 3.1. M is a λ -module. Then M is projective if and only if M is injective.

The proof can be found in [1]

Let $\mathrm{Mod}(\Lambda)$ be the category of Λ -modules and $\underline{\mathrm{Mod}}(\Lambda)$ be the category of $\mathrm{Mod}(\Lambda)$ quotients, the map factoring through projective-injective modules.

In the following we show $\underline{\mathrm{Mod}}(\Lambda)$, called stable module category, is equipped with an endofunctor

$$\Omega : \underline{\mathrm{Mod}}(\Lambda) \rightarrow \underline{\mathrm{Mod}}(\Lambda), \quad M \mapsto \Omega(M).$$

- For $M \in \mathrm{Mod}(\Lambda)$, there exists a short exact sequence $0 \rightarrow X \rightarrow P \rightarrow M \rightarrow 0$ factor through projective-injective module P . Then we let $\Omega(M) = X$. Existence: Let P be the free module generated by the generators of M . A free module is projective, so P is projective. Since Λ is Frobenius, P is also injective.
- $\Omega(M)$ is well-defined. The requirement of factoring through projective-injective P is to show uniqueness of $\Omega(M)$. Let

$$0 \rightarrow \Omega(M) \xrightarrow{i} P \xrightarrow{p} M \rightarrow 0$$

and

$$0 \rightarrow \Omega(M)' \xrightarrow{i'} P' \xrightarrow{p'} M \rightarrow 0$$

There are two short exact sequences. We then obtain two morphisms f , u as follows:

$$\begin{array}{ccccc} \Omega(M)' & \longrightarrow & P' & \xrightarrow{p'} & M \\ \uparrow \scriptstyle f & & \uparrow \scriptstyle u & & \uparrow \scriptstyle id \\ \Omega(M) & \longrightarrow & P & \xrightarrow{p} & M \end{array}$$

where u is induced by lifting property of projective object and f is induced by universal property of kernel.

Similarly we have

$$\begin{array}{ccccc} \Omega(M)' & \longrightarrow & P' & \xrightarrow{p'} & M \\ \downarrow \scriptstyle g & & \downarrow \scriptstyle v & & \downarrow \scriptstyle id \\ \Omega(M) & \longrightarrow & P & \xrightarrow{p} & M \end{array}$$

Next we show $gf = id_{\Omega(M)}$ in $\underline{Mod}(\Lambda)$ and $fg = id_{\Omega(M)'} is similar.$

For $v_u, id_P : P \rightarrow P$, we have $pv_u = p'u = p$ and $pid_P = p$.

So $p(vu - id_P) = 0$.

By the universal property of the kernel, there exists a unique $h : P \rightarrow \Omega(M)$ such that $ih = vu - id_P$.

$$ih = vu - id_P \implies ihi = vui - id_P i,$$

$$\text{i.e., } [\Omega(M) \xrightarrow{i} P \xrightarrow{ih} P] = vu|_{\Omega(M)} - id|_{\Omega(M)} = gf - id_{\Omega(M)}.$$

Since $gf - id_{\Omega(M)}$ factors through projective-injective P , so $gf = id_{\Omega(M)}$ in $\underline{Mod}(\Lambda)$.

It is straightforward to check that Ω is a functor by universal property.

Before introducing the Tate Ext functor, let us first review the ordinary Ext functor.

Key points: $\text{Ext}^i(A, B) = H^i \text{Hom}(A, I_B^\bullet) = H^i \text{Hom}(P_\bullet^A, B)$ where $0 \rightarrow B \rightarrow I_B^\bullet$ is an injective resolution of B and $P_\bullet^A \rightarrow A \rightarrow 0$ is a projective resolution of A .

Remark 3.2. Here, I provide a way to remember them.

1. The notation $\text{Ext}^i(A, B)$ reminds us to construct a cochain complex and compute its i -th cohomology.
2. Injective resolution has an increasing index while projective resolution has a decreasing index.

$0 \rightarrow ? \rightarrow I^\bullet$ has increasing index so $0 \rightarrow ? \rightarrow I^\bullet$ should lie in the co-variant position, which corresponds to B . Hence $\text{Hom}(A, I_B^\bullet)$ is a cochain complex. Similarly, $P_\bullet \rightarrow ? \rightarrow 0$ has decreasing index, so $0 \rightarrow S \rightarrow P_\bullet \rightarrow$

$? \rightarrow 0$ should lie in contravariant position, which corresponds to A . Hence $\text{Hom}(P_\bullet^A, B)$ is a cochain complex.

From above, we find that the notation $P_\bullet$ and $I^\bullet$ is significant. The subscription $\bullet$ in $P_\bullet$ indicates that P_{bullet} has decreasing index.

We define Hochschild Tate cohomology by the Tate Ext functor. Then we come to the Tate Ext functor.

$$\begin{aligned} \text{Ext}_\Lambda^n(M, N) &:= \underline{\text{Mod}}(\Lambda)_{\Omega^{-1}}^n(M, N) \\ &= \underline{\text{Mod}}(\Lambda)(M, \Omega^{-n}(N)) \\ &= \text{Hom}_{\underline{\text{Mod}}(\Lambda)}(M, \Omega^{-n}(N)) \\ &=: \underline{\text{Hom}}_\Lambda(M, \Omega^{-n}(N)) \\ &= \underline{\text{Hom}}_\Lambda(\Omega^n(M), N). \end{aligned}$$

The second and third equations are by definition, and the last equation is because Ω^n is an equivalence.

Claim: $\underline{\text{Hom}}_\Lambda(\Omega^n(M), N) \cong H^n \text{Hom}_\Lambda(P_\bullet, N)$, where $P_\bullet$ is a two-sided (complete) resolution of M , i.e.,

$$\cdots \rightarrow P_3 \xrightarrow{d_3} P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} P_{-1} \xrightarrow{d_{-1}} \cdots$$

such that

- each P_i is a projective-injective module.
- It is an exact sequence.
- Let $\text{Im}(P_n \rightarrow P_{n-1}) =: B_n$. Then we require $B_0 = M$.

Remark 3.3. Care about differences between Hom_Λ and $\underline{\text{Hom}}_\Lambda$. $\text{Hom}_\Lambda = \text{Hom}_{\text{Mod}(\Lambda)}$ and $\underline{\text{Mod}}(\Lambda) = \text{Hom}_{\underline{\text{Mod}}(\Lambda)}$

- Show $\Omega^n(M) = B_n$. We have:

$$\begin{array}{ccccc} P_{n+1} & \xrightarrow{d_{n+1}} & P_n & \xrightarrow{d_n} & P_{n-1} \\ \downarrow & \nearrow & \downarrow & \nearrow & \\ B_{n+1} & & B_n & & \\ \nearrow & & \searrow & & \\ 0 & & & & 0 \end{array}$$

It is easy to check $0 \rightarrow B_{n+1} \rightarrow P_n \rightarrow B_n \rightarrow 0$ is a short exact sequence.

Pick $n = 0$, we have $0 \rightarrow B_1 \rightarrow P_0 \rightarrow B_0 = M \rightarrow 0$. So $B_1 = \Omega(M)$.

Suppose $\Omega^{n-1}(M) = B_{n-1}$. Then $0 \rightarrow B_n \rightarrow P_{n-1} \rightarrow B_{n-1} = \Omega^{n-1}(M) \rightarrow 0$ shows $B_n = \Omega^n(M)$.

- It suffices to show $\underline{\text{Hom}}_\Lambda(B_n, N) \cong H^n(\text{Hom}_\Lambda(P_\bullet, N))$. Then, by $B_n = \Omega^n(M)$, we prove our claim.

$$\cdots \rightarrow \text{Hom}_\Lambda(P_{n-1}, N) \xrightarrow{-\circ d_n} \text{Hom}_\Lambda(P_n, N) \xrightarrow{-\circ d_{n+1}} \text{Hom}_\Lambda(P_{n+1}, N) \rightarrow \cdots$$

$$H^n(\text{Hom}_\Lambda(P_\bullet, N)) = \frac{\{f : P_n \rightarrow N \mid f d_{n+1} = 0\}}{\{g d_n \mid g : P_{n-1} \rightarrow N\}}.$$

1. **Numerator:** $f d_{n+1} = 0$, so $f|_{\text{Im } d_{n+1}} = f|_{B_{n+1}} = 0$.
 $f : P_n \rightarrow N$ with $f|_{B_{n+1}} = 0$ will induce a unique map $\bar{f} : P_n/B_{n+1} \rightarrow N$ such that the following diagram commutes:

$$\begin{array}{ccc} P_n & \xrightarrow{f} & N \\ \downarrow & \nearrow \exists! \bar{f} & \\ B_n = P_n/B_{n+1} & & \end{array}$$

Hence the Numerator $\cong \text{Hom}_\Lambda(B_n, N)$.

2. **Denominator** $= \{f : P_n \rightarrow N \mid f \text{ factors through } d_n : P_n \rightarrow P_{n-1}\}$.

$$\begin{array}{ccc} P_n & \xrightarrow{f} & N \\ d_n \downarrow & \nearrow \exists g & \\ P_{n-1} & & \end{array}$$

We add B_n to this picture:

$$\begin{array}{ccc} & P_n & \xrightarrow{f} N \\ & \downarrow d_n & \nearrow \exists g \\ B_n & \hookrightarrow P_{n-1} & \end{array}$$

Given f lies in denominator, there exists unique $B_n \rightarrow N$ factors through $B_n \hookrightarrow P_{n-1}$.

Given any $B_n \rightarrow N$ factors through $B_n \hookrightarrow P_{n-1}$, there exists unique f lies in denominator. Hence, denominator $= \{f \in \text{Hom}_\Lambda(B_n, N) \mid f \text{ factors through } B_n \hookrightarrow P_{n-1}\}$.

Claim. Denominator $= \{f \in \text{Hom}_\Lambda(B_n, \Lambda) \mid f \text{ factors through some projective-injective module}\}$

Suppose $B_n \rightarrow \Lambda$ factors through projective-injective Q .

$$\begin{array}{ccc} B_n & \xrightarrow{f} & \Lambda \\ \downarrow & \searrow & \uparrow \\ P_{n-1} & \xrightarrow{g} & Q \end{array}$$

Here $g : P_{n-1} \rightarrow Q$ exists since Q is an injective module. So f factors through $B_n \rightarrow P_{n-1}$.

Suppose $B_n \rightarrow \Lambda$ factors through $B_n \rightarrow P_{n-1}$. Since P_{n-1} is projective-injective, so f factors through some projective-injective modules.

Therefore, $\underline{\text{Ext}}_\Lambda^n(M, N) = H^n \text{Hom}(P_\bullet, N)$. It is a parallel equation to ordinary Ext .

Comparison map between Ext and $\underline{\text{Ext}}$.

Construction 3.4. (Comparison maps) We have an obvious projection $P_\bullet \rightarrow P_{\geq 0}$.

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & P_2 & \longrightarrow & P_1 & \longrightarrow & P_0 & \longrightarrow & P_{-1} & \longrightarrow & P_{-2} & \longrightarrow & \cdots \\ & & \downarrow id & & \downarrow id & & \downarrow id & & \downarrow id & & \downarrow id & & \\ \cdots & \longrightarrow & P_2 & \longrightarrow & P_1 & \longrightarrow & P_0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots \end{array}$$

Then apply $\text{Hom}(-, N)$ and we get:

$$\begin{array}{ccccccccc} \cdots & \leftarrow & \text{Hom}(P_2, N) & \leftarrow & \text{Hom}(P_1, N) & \leftarrow & \text{Hom}(P_0, N) & \leftarrow & \text{Hom}(P_{-1}, N) & \leftarrow & \text{Hom}(P_{-2}, N) & \leftarrow & \cdots \\ & & \downarrow id & & \downarrow id & & \downarrow id & & \downarrow id & & \downarrow id & & \\ \cdots & \leftarrow & \text{Hom}(P_2, N) & \leftarrow & \text{Hom}(P_1, N) & \leftarrow & \text{Hom}(P_0, N) & \leftarrow & \text{Hom}(0, N) & \leftarrow & \text{Hom}(0, N) & \leftarrow & \cdots \end{array}$$

The n^{th} cohomology of the bottom line is $\text{Ext}^n(M, N)$ and the n^{th} cohomology of the top line is $\underline{\text{Ext}}^n(M, N)$. Clearly, $P_\bullet \rightarrow P_{\geq 0}$ induces $\text{Ext}_\Lambda^n(M, N) \rightarrow \underline{\text{Ext}}_\Lambda^n(M, N)$.

- For $n > 0$, this map is an isomorphism.
- For $n = 0$,

$$\text{Ext}_\Lambda^0(M, N) = \ker d_1^* / 0 = \ker d_1^*,$$

$$\underline{\text{Ext}}_\Lambda^0(M, N) = \ker d_1^* / \text{Im } d_0^*.$$

So the comparison map $\text{Ext}_\Lambda^n(M, N) \rightarrow \underline{\text{Ext}}_\Lambda^n(M, N)$ is a surjection.

Definition 3.5. Define stable Hochschild cohomology of Λ with coefficients in a Λ -bimodule M as

$$\underline{\text{HH}}^*(\Lambda, M) := \text{Ext}_{\Lambda^e}^*(\Lambda, M), \quad \text{where } \Lambda^e = \Lambda \otimes_k \Lambda^{\text{op}}.$$

Remark 3.6. The left Λ^e -module is equivalent to the Λ -bimodule. Since Ext_R^* only considers left R -modules, we need Λ^e here.

Proposition 3.7 (Proposition 5.7,[3]). An element $f \in \underline{\text{HH}}^{p,q}(\Lambda, \Lambda(\sigma)) = \underline{\text{HH}}^p(\Lambda, {}_{\sigma^q}\Lambda_1)$ is a unit in $\underline{\text{HH}}^{*,*}(\Lambda, \Lambda(\sigma))$ if and only if $f : \Omega^q(\Lambda) \rightarrow {}_{\sigma^q}\Lambda_1$ is an isomorphism in $\underline{\text{Mod}}(\Lambda^e)$.

By this proposition, we have the following fact that generalizes the case of ordinary Ext .

For any $p > 0$ and $f \in \underline{\text{HH}}^{p,q}(\Lambda, \Lambda(\sigma)) = \underline{\text{Ext}}_\Lambda^p(\Lambda, {}_{\sigma^q}\Lambda_1) = \text{Ext}_\Lambda^p(\Lambda, {}_{\sigma^q}\Lambda_1)$, we can take a representing extension

$$0 \rightarrow {}_{\sigma^q}\Lambda_1 \rightarrow P_{p-1} \rightarrow \cdots \rightarrow P_1 \rightarrow P_0 \rightarrow \Lambda \rightarrow 0$$

with $P_{p-2}, P_{p-3}, \dots, P_0$ projective-injective. f is a unit in $\text{HH}^{*,*}(\Lambda, \Lambda(\sigma))$ if and only if P_{p-1} is also projective-injective.

Proof. Firstly, construction of exact sequence $0 \rightarrow {}_{\sigma^q}\Lambda_1 \rightarrow P_{p-1} \rightarrow \cdots \rightarrow P_1 \rightarrow P_0 \rightarrow \Lambda \rightarrow 0$.

We have shown before that $\text{Mod}(\Lambda)$ has enough projectives. Λ is Frobenius, and thus it also has enough injectives.

So for Λ we can find a projective cover $P_0 \rightarrow \Lambda$.

By definition of Ω , we have short exact sequence $0 \rightarrow \Omega(\Lambda) \rightarrow P_0 \rightarrow \Lambda \rightarrow 0$. Find a projective cover for $\Omega(\Lambda)$, we obtain $0 \rightarrow \Omega^2(\Lambda) \rightarrow P_1 \rightarrow \Omega(\Lambda) \rightarrow 0$. We concatenate two short exact sequences and obtain $0 \rightarrow \Omega^2(\Lambda) \rightarrow P_1 \rightarrow P_0 \rightarrow \Lambda \rightarrow 0$. Repeat above steps we can obtain $0 \rightarrow \Omega^{p-1}(\Lambda) \rightarrow P_{p-2} \rightarrow \cdots \rightarrow P_0 \rightarrow \Lambda \rightarrow 0$.

If we continue above extension, we'll have $0 \rightarrow \Omega^p(\Lambda) \rightarrow P_{p-1} \rightarrow \cdots \rightarrow P_0 \rightarrow \Lambda \rightarrow 0$. Moreover, in general, we cannot obtain ${}_{\sigma^q}\Lambda_1$. So we will pick an extension P_{p-1} here and do not require P_{p-1} to be projective-injective.

$$0 \rightarrow {}_{\sigma^q}\Lambda_1 \rightarrow P_{p-1} \rightarrow P_{p-2} \rightarrow \cdots \rightarrow P_0 \rightarrow \Lambda \rightarrow 0$$

$f \in \text{Ext}_\Lambda^p(\Lambda, {}_{\sigma^q}\Lambda_1) = \underline{\text{Hom}}_\Lambda(\Omega^p(\Lambda), {}_{\sigma^q}\Lambda_1)$. So f meets the requirement in [Proposition 5.7,[3]]. Therefore, f is a unit if and only if $f : \Omega^p(\Lambda) \rightarrow {}_{\sigma^q}\Lambda_1$ is an isomorphism.

Suppose f is an isomorphism.

So $0 \rightarrow {}_{\sigma^q}\Lambda_1 \rightarrow P_{p-1} \rightarrow P_{p-2} \rightarrow \cdots \rightarrow P_0 \rightarrow \Lambda \rightarrow 0$

$$0 \rightarrow \Lambda \rightarrow P_{p-1} \rightarrow P_{p-2} \rightarrow \cdots \rightarrow P_0 \rightarrow \Lambda \rightarrow 0$$

Conversely suppose P_{p-1} is projective-injective.

Consider the following diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & {}_{\sigma^q}\Lambda_1 & \hookrightarrow & P_{p-1} & \xrightarrow{d_{p-1}} & P_{p-2} \longrightarrow \cdots \longrightarrow \Lambda \longrightarrow 0 \\ & & & \nearrow & & \searrow & \uparrow \\ & & \Omega^p(\Lambda) & & & & \Omega^{p-1}(\Lambda) \end{array}$$

By exactness, $\Omega^p(\Lambda) \cong \ker d_{p-1} = {}_{\sigma^q}\Lambda_1$.

□

References

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